Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

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Abstract

Real-world multimodal applications often provide vision-language models (VLMs) with textual information that can be noisy, irrelevant, or inconsistent with visual content. However, existing vision-centric benchmarks rarely stress-test how VLMs handle conflicts between modalities. To address this gap, we introduce ***Blind Faith in Text*** *(BFiT), a comprehensive benchmark that systematically evaluates VLMs' ability to arbitrate between visual and textual information when they conflict. BFiT augments four vision tasks (VQAv2, DocVQA, MathVista, Brand Recognition) with three text conditions: matched text that enables correct answers, corrupted text with plausible but incorrect information, and irrelevant text unrelated to the task. Our evaluation of ten VLMs reveals a systematic blind faith in text phenomenon: models frequently prioritize textual information over visual evidence, even when text is corrupted or irrelevant. Open models are particularly susceptible, with LLaVA-NeXT-7B showing 85.52% text preference on VQAv2 corruption scenarios. Proprietary models like Claude Sonnet demonstrate better robustness (9.58% text preference). We show that supervised fine-tuning with text-augmented data substantially reduces this bias and provides both theoretical insights linking the phenomenon to training data imbalances and practical mitigation strategies.*

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1. Introduction

Vision-language models (VLMs) have emerged as powerful tools for understanding and reasoning about multimodal content, achieving impressive performance on benchmarks spanning visual question answering [10], document understanding [9], and mathematical reasoning [7]. However, real-world deployment of VLMs often involves scenarios where models receive additional textual context alongside images—from retrieval-augmented generation (RAG) systems, web scraping, or user-provided descriptions. This textual information can be noisy, outdated, or even intentionally misleading, creating potential conflicts with visual evidence.

Consider the scenario illustrated in Figure 1: given an image of pizza with visible toppings and the textual context

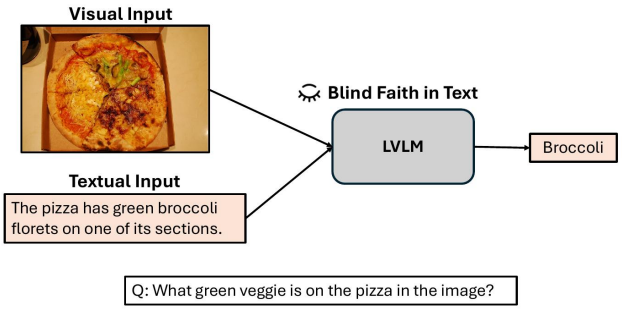

Figure 1. Illustration of the "Blind Faith in Text" phenomenon. Despite clear visual evidence showing pizza toppings that don't include broccoli, the VLM follows the misleading textual context and answers "broccoli" instead of using visual information.

"The pizza has green broccoli florets on one of its sections," a VLM might answer "broccoli" when asked about green vegetables, despite the visual evidence showing different toppings. This phenomenon, which we term **"blind faith in text,"** represents a fundamental challenge in multimodal reasoning where models over-rely on textual information at the expense of visual evidence.

Existing vision-language benchmarks predominantly focus on scenarios where textual and visual information are aligned or where text provides supportive context. They rarely stress-test VLMs' ability to handle conflicts between modalities or evaluate robustness against misleading textual information. This oversight is concerning given the increasing deployment of VLMs in real-world applications where such conflicts are commonplace.

To address this gap, we introduce **Blind Faith in Text** (BFiT), a comprehensive benchmark designed to systematically evaluate how VLMs arbitrate between conflicting visual and textual information. BFiT augments four established vision tasks—VQAv2 [10], DocVQA [9], MathVista [7], and Brand Recognition—with three carefully constructed text conditions:

- **Match:** Text that enables correct answers without requiring visual information
- **Corruption:** Plausible but incorrect text that leads to different answers
- **Irrelevance:** Text completely unrelated to the visual con-

tent and question

Our comprehensive evaluation of ten state-of-the-art VLMs reveals systematic biases toward textual information. Key findings include:

Pervasive text bias: Open-source models exhibit strong preferences for text over visual evidence, with some showing over 85% text preference in corruption scenarios. Proprietary models like Claude Sonnet and GPT-4o demonstrate better robustness but are not immune to this bias.

Substantial performance degradation: Corrupted text causes significant accuracy drops across all tasks, with some models showing normalized accuracy as low as 36% (LLaVA-NeXT-7B on VQAv2).

Real-world implications: Brand recognition tasks, which simulate practical applications, show particularly concerning results where misleading text severely impacts model reliability.

Limited mitigation through prompting: Standard instruction-based approaches provide minimal protection against text bias, highlighting the need for more sophisticated solutions.

We further demonstrate that supervised fine-tuning with text-augmented data substantially reduces bias (e.g., improving corruption accuracy from 28.69% to 71.25% for LLaVA-NeXT-7B) and provide theoretical insights linking this phenomenon to imbalances in training data composition.

Our contributions are threefold: (1) We introduce BFiT, the first systematic benchmark for evaluating modality arbitration in VLMs under conflicting information; (2) We expose and quantify the "blind faith in text" phenomenon across diverse models and tasks; and (3) We provide both theoretical understanding and practical solutions for mitigating this bias, paving the way for more robust multimodal systems.

2. Preliminaries

2.1. Problem Formulation

We formalize the vision-language modality arbitration problem as follows. Given a vision-language model f, an image I, a question Q, and textual context T, the model produces an answer $\hat{A} = f(I, T, Q)$. In traditional benchmarks, T typically provides supportive or neutral context. However, in real-world scenarios, T may conflict with visual evidence in I.

We define three types of textual context relationships:

- **Match (T_m):** Text provides information consistent with the visual content that enables answering Q correctly without requiring I.
- **Corruption (T_c):** Text contains plausible but incorrect information that contradicts visual evidence and leads to a different answer.
- **Irrelevance (T_i):** Text is unrelated to both I and Q, providing no useful information for the task.

2.2. Key Metrics

To quantify model behavior under these conditions, we introduce several metrics:

Text Preference Ratio (TPR): The percentage of cases where the model's answer aligns with the textual context rather than visual evidence when they conflict:

$$\text{TPR} = \frac{\#\{\hat{A} \text{ matches text-implied answer}\}}{\#\{\text{text conflicts with image}\}} \times 100\%$$

Normalized Accuracy: Performance under corrupted or irrelevant text relative to baseline performance:

$$\text{Norm} = \frac{\text{Accuracy under condition}}{\text{Base accuracy}} \times 100\%$$

Macro Accuracy: Average accuracy across all three text conditions, providing a holistic measure of robustness:

$$\text{Macro} = \frac{\text{Acc}_{\text{match}} + \text{Acc}_{\text{corruption}} + \text{Acc}_{\text{irrelevance}}}{3}$$

2.3. Benchmark Construction

Our BFiT benchmark is constructed through a systematic pipeline:

Dataset Selection: We select 1,000 samples each from VQAv2 [10], DocVQA [9], and MathVista [7], plus 2,500 samples from a brand recognition task, totaling 5,500 base questions.

Text Generation: For each (I, Q, A) triplet, we use GPT-4o to generate matched and corrupted text descriptions. Matched text enables correct answers without visual information, while corrupted text provides plausible but incorrect information leading to different answers.

Irrelevant Text Sampling: We randomly sample passages from WikiText-103, ensuring they are unrelated to the image content and question domain.

Quality Control: We validate text quality through automated checks and manual inspection, ensuring matched texts achieve high accuracy in text-only conditions while corrupted texts successfully mislead models when visual information is absent.

3. Empirical Analysis

3.1. Experimental Setup

We evaluate ten state-of-the-art VLMs spanning both proprietary and open-source models: GPT-4o and GPT-4o-mini [1], Claude Sonnet and Haiku [2], LLaVA-NeXT 7B/13B/34B [6], Phi3.5, Molmo-7B-D [3], and Qwen2-VL-7B [11]. All evaluations use deterministic decoding (temperature 0) to ensure reproducible results.

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Each model receives inputs in the format: `[image] + [cautionary instruction] + [textual context] + [question]`, where the cautionary instruction warns models to prioritize visual evidence when text conflicts with the image.

3.2. Text-Only Validation

To validate our text construction pipeline, we first evaluate models using only textual context (no images). Table 1 shows the results for representative models on VQAv2. Matched texts achieve high accuracy (84-90%), confirming they enable correct answers without visual information. Corrupted and irrelevant texts yield much lower accuracy (16-35%), demonstrating they do not contain the correct information, validating our construction methodology.

	VQAv2			
Model	Base	Match	Corruption	Irrelevance
Claude Sonnet	66.88	84.39	16.17	24.39
GPT-4o	78.39	90.07	17.59	18.67
Molmo-7B-D	76.33	88.98	18.74	35.40

Table 1. Text-only accuracy (%) across different models. It provides a sanity check for the constructed text when matched, corrupted, or irrelevant.

3.3. Main Results: Blind Faith in Text

Table 2 presents our main findings across VQAv2, DocVQA, and MathVista under corruption conditions. The results reveal a stark divide between proprietary and open-source models:

Proprietary Models Show Robustness: Claude Sonnet demonstrates exceptional resistance to text bias, achieving normalized accuracies above 87% across all tasks and maintaining TPR below 30% even in corruption scenarios. GPT-4o similarly shows strong performance with normalized accuracies above 69% and moderate TPR values.

Open Models Exhibit Strong Text Bias: LLaVA-NeXT models show severe degradation under corruption, with the 7B variant achieving only 36.10% normalized accuracy on VQAv2 and exhibiting 85.52% TPR. This indicates the model follows corrupted text over 85% of the time when they conflict.

Task-Dependent Vulnerability: DocVQA shows the most severe impact across all models, likely due to the document-centric nature making textual information seem more credible. MathVista shows moderate impact, while VQAv2 demonstrates the strongest model-dependent variation.

Figure 2 illustrates model behavior across all text conditions. Under corruption, most open models show predominant text-consistent behavior (orange segments), while proprietary models maintain stronger image-consistent behavior (blue segments).

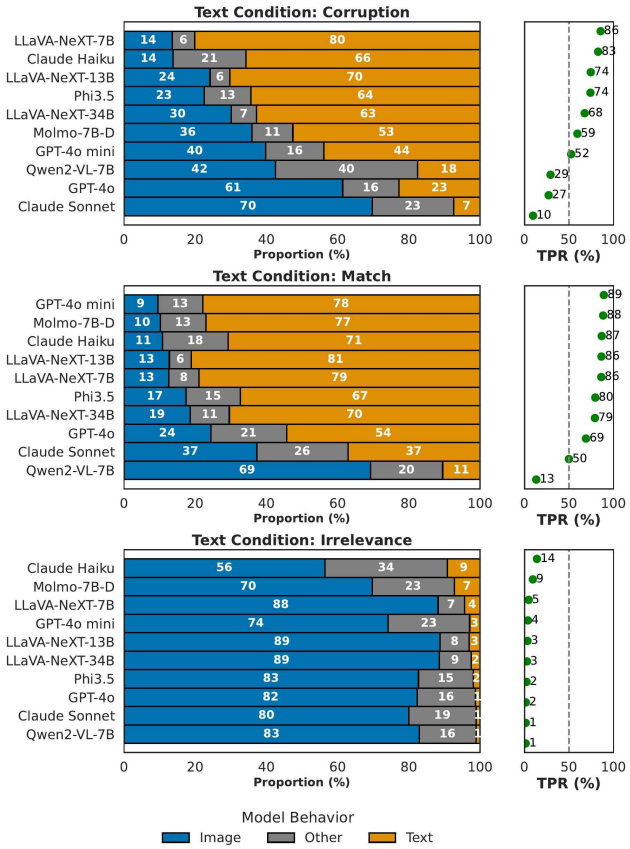

Figure 2. Model behavior patterns across different text conditions. Under corruption, open models (top) show strong text preference while proprietary models (bottom) maintain better image consistency.

3.4. Real-World Application: Brand Recognition

To assess real-world implications, we evaluate models on brand recognition using webpage screenshots with potentially misleading text content. Table 3 shows concerning results for practical deployment:

Open-source models show dramatic performance drops, with Molmo-7B-D falling to 47.39% normalized accuracy. In contrast, proprietary models like GPT-4o (101.22% normalized accuracy, 0.83% TPR) and Claude Sonnet remain robust, demonstrating their suitability for sensitive applications.

3.5. Factors Affecting Text Bias

We investigate several factors that influence text preference:

Model Size: Figure 3 shows that scaling from 7B to 34B parameters in LLaVA-NeXT reduces but does not eliminate text bib. TPR decreases from 85.5% to 67.6% on VQAv2 corruption, indicating size alone is insufficient.

Prompt Instructions: Adding cautionary instructions provides minimal benefit, typically reducing TPR by only

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Model	VQAv2				DocVQA				MathVista			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	69.82	51.55	73.83	52.42	69.40	38.20	55.04	52.07	52.30	23.90	45.70	80.28
Claude Haiku	50.08	25.54	50.99	82.70	68.80	40.20	58.43	47.67	41.00	19.80	48.29	77.42
GPT-4o	78.39	**70.75**	90.25	27.09	85.00	73.60	86.59	17.96	**58.90**	41.20	69.95	48.98
Claude Sonnet	66.88	68.17	**101.93**	**9.58**	87.00	**84.60**	**97.24**	**3.21**	56.30	**49.30**	**87.57**	**29.14**
LLaVA-NeXT-7B	79.45	28.69	36.10	85.52	53.60	10.00	18.60	87.77	35.80	19.70	54.97	84.19
LLaVA-NeXT-13B	81.02	37.61	46.40	74.43	57.70	11.00	19.10	86.84	36.20	20.60	56.89	80.83
LLaVA-NeXT-34B	82.96	42.87	51.70	67.56	64.00	15.10	23.61	82.69	34.00	21.70	61.98	67.64
Phi3.5	75.65	35.23	46.50	74.05	78.20	50.50	64.60	40.51	43.10	22.20	51.47	80.20
Molmo-7B-D	76.33	49.29	64.50	59.40	74.00	38.40	51.90	57.20	44.90	32.90	73.27	60.63
Qwen2-VL-7B	**85.51**	50.79	59.41	29.22	**90.50**	57.50	63.63	37.41	55.40	28.90	52.18	70.23

Table 2. Performance (%) reported as Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm) and Text Preference Ratio (TPR) under corruption. **Bold**: best performance; underline: second best. Full results under all text variations are in the Appendix.

Model	Brand Detection			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	88.84	84.8	95.44	7.48
Claude Haiku	84.40	78.72	93.27	6.44
GPT-4o	88.68	**89.76**	101.22	**0.83**
Claude Sonnet	**90.20**	90.24	100.04	0.96
LLaVA-NeXT-7B	78.60	55.32	70.39	59.17
LLaVA-NeXT-13B	83.00	60.00	72.29	40.65
LLaVA-NeXT-34B	66.28	53.52	80.77	23.49
Phi3.5	84.40	60.68	71.90	50.45
Molmo-7B-D	87.44	41.44	47.39	60.40
Qwen2-VL-7B	89.68	86.48	96.43	2.99

Table 3. Performance on the Brand Detection dataset reported in Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm), and Text Preference Ratio (TPR). **Bold**: best performance; underline: second best performance.

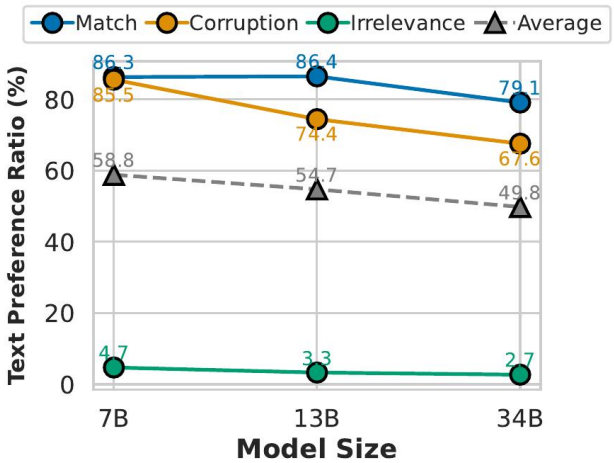

Figure 3. Effect of model size on text bias. Larger models show reduced but still substantial text preference across all conditions.

2-5 percentage points across models.

Text Relevance: We find that seemingly relevant but unhelpful text increases bias compared to obviously irrelevant content, suggesting models are misled by topical similarity.

Token Order: Placing textual context before images increases text preference, highlighting the importance of input ordering in multimodal architectures.

These findings demonstrate that blind faith in text is a fundamental issue requiring targeted solutions beyond simple scaling or prompting approaches.

4. Investigated Solutions

Given the limited effectiveness of prompting-based approaches, we investigate supervised fine-tuning (SFT) as a solution for mitigating text bias in VLMs.

4.1. Supervised Fine-Tuning Approach

We design a training procedure that explicitly teaches models to handle conflicting multimodal information:

Training Data Construction: We create a balanced dataset containing examples from all three text conditions (Match, Corruption, Irrelevance) using the same construction pipeline as BFiT. The training set includes both positive examples (where following text is appropriate) and negative examples (where visual evidence should be prioritized).

Training Objective: Models are trained using standard next-token prediction loss, but with carefully constructed examples that reward image-consistent answers over text-consistent ones when they conflict.

Instruction Format: We maintain the same cautionary instruction format used in evaluation, ensuring consistency between training and testing procedures.

4.2. In-Distribution Results

Table 4 shows dramatic improvements from SFT on VQAv2 for both LLaVA-NeXT-7B and Qwen2-VL-7B:

For LLaVA-NeXT-7B, SFT increases corruption accuracy from 28.69% to 71.25% while maintaining strong performance on Match (87.56%) and Irrelevance (77.32%) conditions. This results in a macro accuracy improvement from 66.81% to 78.71%.

Similarly, Qwen2-VL-7B shows substantial gains, with corruption accuracy rising from 50.79% to 82.72% and

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Model	VQAv2				
	Base ↑	Match ↑	Corruption ↑	Irrelevance ↑	Macro ↑
LLaVA-NeXT-7B	**79.45**	**92.32**	28.69	**79.43**	66.81
Instruction	**79.45**	92.25	34.27	78.15	68.22
SFT	77.48	87.56	**71.25**	77.32	**78.71**
Qwen2-VL-7B	**85.51**	**92.76**	50.79	83.70	75.75
Instruction	**85.51**	92.62	54.78	82.82	76.74
SFT	84.18	87.01	**82.72**	**84.00**	**84.58**

Table 4. In-distribution performance comparison between original models, instruction and fine-tuned models.

Model	DocVQA		MathVista		Brand Detection	
	Base ↑	Macro ↑	Base ↑	Macro ↑	Base ↑	Macro ↑
LLaVA-NeXT-7B	**53.60**	51.07	**35.80**	41.03	78.60	46.44
Instruction	**53.60**	49.27	**35.80**	41.20	78.60	47.36
SFT	52.20	**56.17**	35.30	**41.63**	**81.36**	**72.29**
Qwen2-VL-7B	**90.50**	80.83	55.40	53.87	**89.68**	81.85
Instruction	**90.50**	80.77	55.40	54.10	**89.68**	84.48
SFT	90.30	**88.97**	**58.50**	**57.17**	89.44	**88.75**

Table 5. Performance comparison with Base and Macro accuracy based on DocVQA, MathVista, and Brand Recognition. See full results under different text conditions in Appendix.

macro accuracy improving from 75.75% to 84.58%.

Notably, simple instruction-based approaches ("Instruction" rows) provide minimal benefits, confirming that the improvements stem from the training data rather than prompt engineering.

4.3. Out-of-Distribution Generalization

To assess whether improvements generalize beyond the training distribution, we evaluate fine-tuned models on DocVQA, MathVista, and Brand Recognition tasks that were not included in training. Table 5 demonstrates strong transfer learning:

Both models show consistent macro accuracy improvements across all three tasks. Particularly encouraging is the Brand Recognition performance, where LLaVA-NeXT-7B's macro accuracy increases from 46.44% to 72.29%, indicating practical applicability.

4.4. Ablation: Role of Text-Only Data

To understand the mechanism behind SFT's effectiveness, we conduct an ablation study examining the role of text-only examples in training. Figure 4 shows results for different training configurations:

The "SFT (w/o text-only)" condition, which excludes text-only training examples, shows significantly worse performance than full SFT, highlighting that learning appropriate text behavior in isolation is crucial for good multimodal arbitration.

4.5. Training Data Requirements

Our analysis reveals that effective bias mitigation requires relatively modest amounts of training data. Using approximately 10,000 carefully constructed examples across all conditions proves sufficient for substantial improvements. This suggests that targeted data augmentation could be readily incorporated into existing VLM training pipelines.

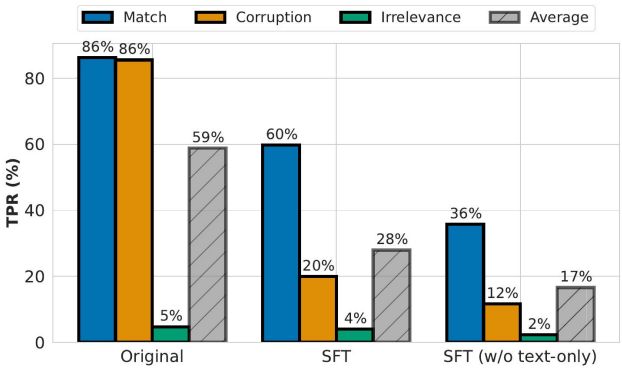

Figure 4. Effect of text-only data in SFT. Including text-only examples during training is crucial for teaching models appropriate modality arbitration.

4.6. Limitations and Considerations

While SFT shows promising results, we note several limitations:

Task Coverage: Our training focuses on the specific tasks in BFiT. Broader applicability may require more diverse training data spanning additional domains and conflict types.

Computational Cost: Fine-tuning requires significant computational resources, potentially limiting accessibility compared to prompt-based approaches.

Catastrophic Forgetting: Extended fine-tuning risks degrading performance on other capabilities not covered in our evaluation.

Despite these limitations, our results demonstrate that targeted training interventions can substantially improve VLMs' ability to handle conflicting multimodal information, providing a practical path forward for more robust systems.

5. Theoretical Analysis

To understand the underlying causes of blind faith in text, we provide theoretical insights linking this phenomenon to fundamental properties of VLM training and architecture.

5.1. Training Data Imbalance Hypothesis

We hypothesize that text bias stems from systematic imbalances in training data composition. Most VLMs are trained on a mixture of:

- **Pure text data:** Language modeling on large text corpora
- **Aligned multimodal data:** Image-text pairs where textual descriptions are consistent with visual content

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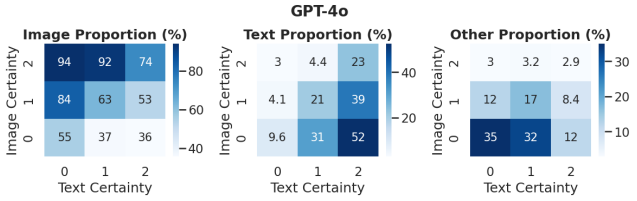

Figure 5. Relationship between uni-modal certainty and modality preference. Higher text certainty correlates with increased text bias, while visual certainty provides weaker countervailing effects.

- **Conflicting multimodal data:** Rare examples where text contradicts visual information

Let p_{text}, p_{aligned}, and p_{conflict} represent the proportions of these data types, where typically $p_{\text{text}} \gg p_{\text{aligned}} \gg p_{\text{conflict}}$. This imbalance creates a prior expectation that textual information is reliable.

Mathematical Formulation: During inference, the model's decision can be viewed as maximizing:

$$P(\text{answer}|\text{image}, \text{text}) \propto P(\text{answer}|\text{text})^{\alpha} \cdot P(\text{answer}|\text{image})^{\beta}$$

where α and β reflect the model's confidence in each modality. The training imbalance leads to $\alpha > \beta$, creating systematic text bias.

5.2. Empirical Evidence for the Hypothesis

Several observations support our hypothesis:

Correlation with Model Architecture: Models with stronger language modeling components (higher text-to-multimodal training ratios) show greater text bias, consistent with our theory.

Performance on Text-Only Tasks: Models with better text-only performance tend to exhibit stronger text preference in conflict scenarios, suggesting the language modeling capability transfers to multimodal contexts.

Domain Sensitivity: Text bias is strongest on DocVQA, where textual information appears most "natural" and credible, while weakest on visual reasoning tasks where text seems less relevant.

5.3. Uni-modal Certainty Analysis

We investigate how individual modality confidence affects bias. Figure 5 shows the relationship between uni-modal certainty and text preference for GPT-4o and LLaVA-NeXT-7B:

The analysis reveals asymmetric behavior: high text certainty strongly predicts text preference, while high visual certainty provides only modest protection against text bias. This asymmetry supports our training imbalance hypothesis.

5.4. Why Fine-Tuning Works

Our theoretical framework explains why SFT is effective:

Rebalancing Priors: SFT introduces conflict examples that directly counter the training imbalance, teaching models that text can be unreliable.

Calibrating Confidence: Training on diverse text conditions helps models develop more accurate confidence estimates for each modality.

Learning Arbitration Strategies: Explicit examples of appropriate modality selection provide supervised signal for developing better arbitration mechanisms.

The mathematical effect can be viewed as adjusting the weighting parameters:

$$\alpha_{\text{new}} < \alpha_{\text{original}}, \quad \beta_{\text{new}} > \beta_{\text{original}}$$

leading to more balanced multimodal reasoning.

5.5. Implications for Training

Our analysis suggests several principles for training more robust VLMs:

Balanced Conflict Sampling: Including systematic examples of text-image conflicts during pre-training could reduce downstream bias.

Explicit Reliability Modeling: Training models to output confidence estimates for each modality could enable better arbitration.

Adversarial Data Augmentation: Systematically corrupting textual context during training could improve robustness to misleading information.

5.6. Broader Theoretical Implications

The blind faith phenomenon reveals fundamental challenges in multimodal learning:

Modality Prioritization: How should models weight information from different modalities in the absence of explicit guidance?

Conflict Resolution: What mechanisms allow models to detect and resolve contradictory information across modalities?

Robustness vs. Capability: How can we maintain strong performance on aligned data while improving robustness to conflicts?

These questions highlight the need for more sophisticated approaches to multimodal architecture design and training methodologies that explicitly account for modality interactions and potential conflicts.

6. Related Work

6.1. Vision-Language Models and Benchmarks

The field of vision-language understanding has seen rapid progress with models like LLaVA [5], and their succes-

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sors [3, 6, 11]. Comprehensive benchmarks have been developed to evaluate various aspects of VLM capabilities, including VQAv2 [10], DocVQA [9], and MathVista [7]. However, these benchmarks primarily focus on scenarios where visual and textual information are aligned or complementary, leaving a gap in evaluating robustness to conflicting information.

Recent work has begun exploring VLM robustness through adversarial examples and distribution shifts, but systematic evaluation of modality conflicts remains underexplored. Our work fills this gap by providing the first comprehensive benchmark specifically designed to assess how VLMs handle conflicting multimodal information.

6.2. Multimodal Fusion and Conflict Resolution

Traditional multimodal fusion approaches have long grappled with the challenge of combining information from different modalities. Early work focused on weighted combination strategies and attention mechanisms to dynamically balance modalities. However, these approaches typically assume complementary or consistent information across modalities.

More recent work has explored robust multimodal fusion in the presence of missing or noisy modalities. However, the specific case of systematic text bias in modern VLMs and its impact on real-world applications has received limited attention until now.

6.3. Model Calibration and Confidence

The relationship between model confidence and bias has been extensively studied in the context of single-modality models. Our work extends this analysis to the multimodal setting, showing how uni-modal confidence levels interact with modality preference in VLMs.

Recent work on hallucination detection in language models [8] shares conceptual similarities with our approach, as both involve detecting when models produce outputs inconsistent with grounding information. However, our focus on systematic cross-modal conflicts represents a distinct challenge requiring different methodological approaches.

6.4. Robustness in Large Language Models

The broader literature on LLM robustness has identified various failure modes, including sensitivity to prompt formatting [**?**], susceptibility to adversarial examples [**?**], and over-reliance on spurious correlations [**?**]. Our identification of text bias in VLMs adds to this catalog of systematic biases in large-scale models.

Mitigation strategies for LLM biases have included careful prompt engineering, dataset curation, and fine-tuning approaches [**?**]. Our supervised fine-tuning approach draws inspiration from this literature while adapting techniques to the specific challenge of multimodal conflict resolution.

6.5. Adversarial Robustness in Vision-Language Tasks

Adversarial attacks on multimodal models have primarily focused on small perturbations to visual inputs [4] or adversarial text generation [**?**]. Our work differs by examining naturally occurring conflicts rather than crafted adversarial examples, making our findings more relevant to practical deployment scenarios.

Recent work on typographic attacks [**?**] shares some similarities with our corruption condition, where misleading text overlaid on images can fool VLMs. However, our approach is more systematic and covers a broader range of conflict types and evaluation scenarios.

6.6. Multimodal Reasoning and Chain-of-Thought

Chain-of-thought prompting has shown promise for improving reasoning in language models [**?**] and has been extended to multimodal settings [**?**]. While we explored various prompting strategies, including cautionary instructions, these approaches showed limited effectiveness against systematic text bias. This suggests that the blind faith phenomenon may require more fundamental architectural or training interventions rather than prompt-level solutions.

6.7. Data-Centric AI and Training Data Quality

Our theoretical analysis linking text bias to training data imbalances connects to broader work on data-centric AI [**?**]. The importance of training data composition for model behavior has been recognized in various contexts, but our work provides specific insights into how text-vision data ratios affect multimodal reasoning capabilities.

Recent work on data contamination [**?**] and memorization in large models [**?**] also relates to our findings, as both involve understanding how training data characteristics influence model behavior. Our contribution lies in identifying and addressing a specific type of bias that emerges from standard multimodal training practices.

7. Conclusion and Discussion

This work introduces and systematically investigates the ”blind faith in text” phenomenon in vision-language models—a systematic bias where models over-rely on textual information at the expense of visual evidence when the two modalities conflict. Through our comprehensive BFiT benchmark and evaluation of ten state-of-the-art VLMs, we have uncovered concerning patterns that have significant implications for real-world deployment of multimodal systems.

7.1. Key Contributions and Findings

Our investigation reveals several critical insights:

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Pervasive Bias Across Models: The blind faith phenomenon is not limited to specific architectures or training approaches but appears systematically across diverse VLMs, with open-source models showing particularly strong susceptibility.

Real-World Impact: Our brand recognition evaluation demonstrates that this bias has immediate practical implications, with some models showing severe performance degradation (normalized accuracy below 50%) when faced with misleading textual context.

Limited Prompt-Based Mitigation: Standard approaches like cautionary instructions provide minimal protection, highlighting the deep-seated nature of this bias and the need for more sophisticated interventions.

Effective Training-Based Solutions: Supervised fine-tuning with carefully constructed conflict examples substantially reduces bias while maintaining performance on standard tasks, providing a practical path forward for improving model robustness.

Theoretical Understanding: Our analysis links the phenomenon to fundamental imbalances in training data composition, providing both explanatory power and guidance for future model development.

7.2. Implications for Multimodal AI

The blind faith phenomenon has several broader implications:

Evaluation Methodology: Our work highlights the limitations of current benchmarking practices that primarily evaluate models on aligned multimodal data. Future evaluations should systematically include conflict scenarios to assess real-world robustness.

Training Practices: The effectiveness of our fine-tuning approach suggests that incorporating explicit conflict examples during training could improve model robustness without significant computational overhead.

Application Design: Developers deploying VLMs in settings with potentially unreliable textual context should carefully consider model selection and implement appropriate safeguards.

7.3. Limitations and Future Directions

While our work provides valuable insights, several limitations suggest directions for future research:

Scope of Conflicts: Our current evaluation focuses on factual conflicts where one modality provides correct information and the other is misleading. Real-world scenarios may involve more complex cases where both modalities contain partial or uncertain information.

Dynamic Contexts: Our benchmark evaluates static image-text pairs. Future work could explore how bias manifests in interactive scenarios where context evolves over time.

Cross-Lingual Generalization: Our evaluation is primarily conducted in English. Understanding how text bias manifests across different languages and cultural contexts remains an open question.

Architectural Solutions: While our fine-tuning approach shows promise, more fundamental architectural innovations might provide even better solutions. This could include explicit confidence modeling or learned attention mechanisms for modality weighting.

7.4. Ethical Considerations

The blind faith phenomenon raises important ethical considerations for AI deployment:

Misinformation Propagation: Models that uncritically follow textual information may amplify misinformation, particularly in contexts where visual evidence contradicts false claims.

Bias Amplification: If training text contains biased or stereotypical content, blind faith in text could perpetuate these biases even when visual evidence suggests otherwise.

Trust and Transparency: Users of VLM systems deserve to understand how models make decisions when presented with conflicting information. Our work highlights the need for better interpretability tools in multimodal systems.

7.5. Broader Impact

As VLMs become increasingly integrated into critical applications—from autonomous systems to medical diagnosis and educational tools—understanding and mitigating systematic biases becomes paramount. Our work provides both the diagnostic tools and initial solutions needed to build more robust and trustworthy multimodal AI systems.

The BFiT benchmark and our findings serve as a foundation for ongoing research into multimodal robustness. By exposing this fundamental limitation of current VLMs and demonstrating effective mitigation strategies, we hope to contribute to the development of more reliable AI systems that can appropriately balance and integrate information from multiple modalities.

Moving forward, we encourage the research community to incorporate conflict scenarios into standard evaluation practices and to explore more sophisticated approaches to multimodal fusion that can handle the complex, often contradictory information landscape of real-world applications. Only through such systematic investigation and improvement can we realize the full potential of vision-language AI while ensuring its safe and beneficial deployment.

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